Chapter 12 · Semester 1

HEAT TRANSFER

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4 worked examples 2 simulations Exercises with answers

Heat

Heat

is the **energy transferred from one body to another because of a temperature difference**.

- Heat always flows from a region of higher temperature to a region of lower temperature, until thermal equilibrium is reached.
- Heat is a scalar; unit joule (J).
- Heat is transferred by three mechanisms: **conduction**, **convection** and **radiation**.

12.1 Conduction

Conduction

is the process by which **heat is transferred through a substance from a region of high temperature to a region of lower temperature**, without any movement of the substance as a whole.

Mechanism of conduction

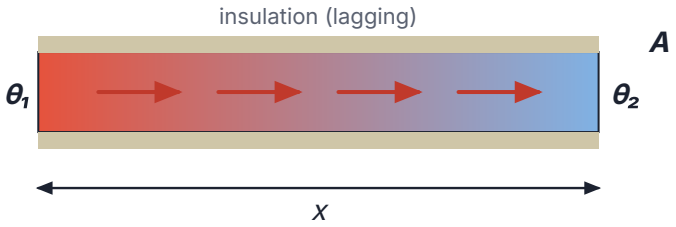
In a non-metal (insulator) rod

- At the hot end A, the molecules receive heat and vibrate faster, with larger amplitude.
- They collide with neighbouring, colder molecules and pass energy to them.
- The energy is passed along molecule by molecule (lattice vibrations) until the cold end B becomes hot.

In a metal rod

- A metal has many **free electrons**.
- At the heated end A the free electrons gain thermal energy and move faster.
- They travel quickly through the metal and collide with atoms in the colder regions, transferring energy to them (in addition to the lattice vibrations).
- Free electrons are light and fast, so this transfer is much quicker: **metals conduct heat much faster than insulators**.

Rate of heat flow



A lagged rod of cross-sectional area A and length x, ends at $\theta_1 > \theta_2$. No heat is lost from the sides.

Steady state

is reached when the **temperature at every point of the rod no longer changes with time**; the rate of heat flow is then the same through every cross-section.

The rate of heat flow through the rod depends on (a) the cross-sectional area A, (b) the temperature gradient, and (c) the material of the rod (e.g. copper or iron).

Temperature gradient

is the **change in temperature per unit length** along the direction of heat flow: $\frac{d\theta}{dx}$ (unit K m^{-1} or °C m^{-1}).

$$\frac{dQ}{dt} \propto A\frac{d\theta}{dx} \Rightarrow \frac{dQ}{dt} = -kA\frac{d\theta}{dx}$$

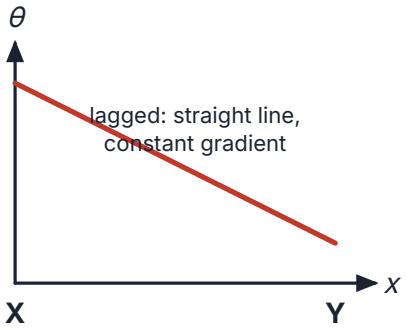
the minus sign: heat flows in the direction of decreasing temperature. Unit of the rate of heat flow: J s^{-1} = W

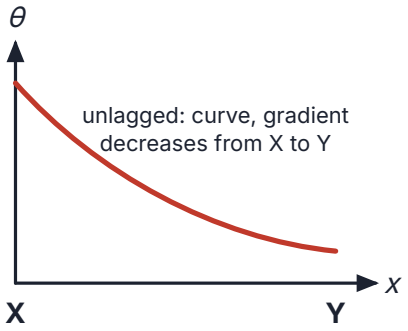
Thermal conductivity, k

is the **rate of heat flow per unit cross-sectional area per unit temperature gradient**, perpendicular to the area: $k = -\frac{dQ/dt}{A(d\theta/dx)}$. Unit W m^{-1} K^{-1}. Good conductors have large k.

Substance	k / W m^{-1} K^{-1}
Silver	406.0
Copper	385.0
Steel	50.2
Glass	0.8
Wood	0.08

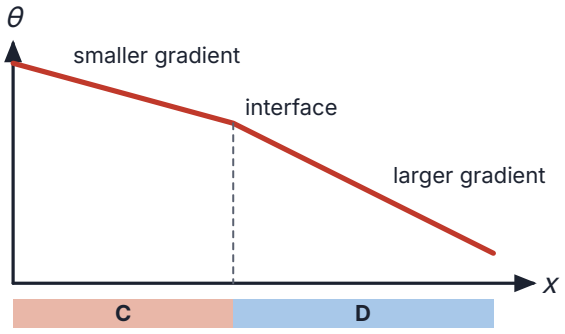
Insulated (lagged) and non-insulated (unlagged) rods





- **Unlagged** bar XY: heat is lost from the sides, so less heat reaches the far end. The rate of heat flow decreases along the bar and the temperature gradient decreases from X to Y: the θ–x graph is a **curve**.
- **Lagged** bar: no heat is lost from the sides. The rate of heat flow is the same everywhere and, with A and k constant, the temperature gradient is **constant**: the θ–x graph is a **straight line**, $\frac{d\theta}{dx} = \frac{\theta_2 - \theta_1}{x}$.

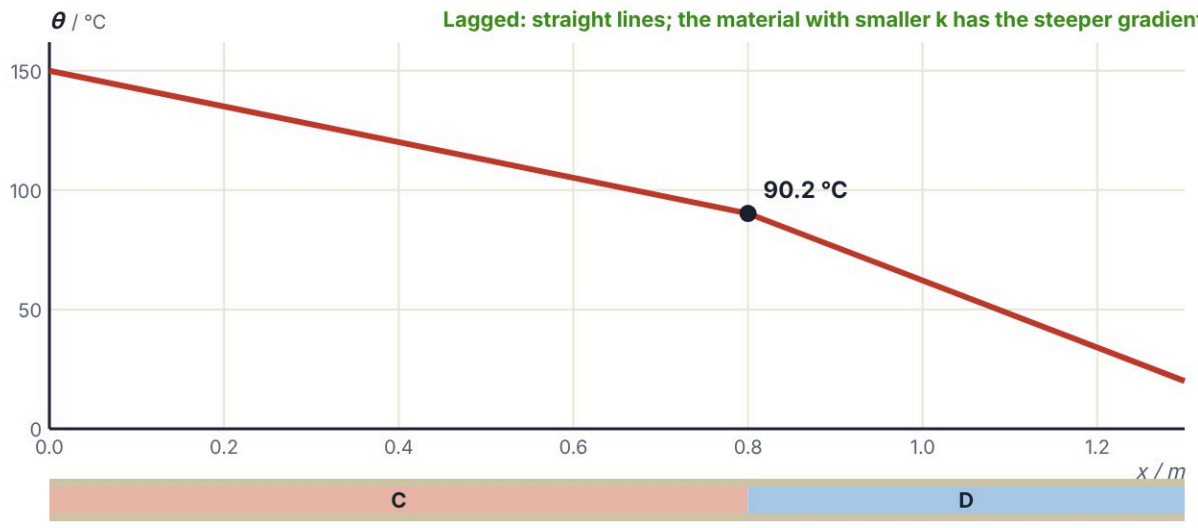
Heat conduction through layers of different materials



- In a lagged composite bar of materials C and D at steady state, the **rate of heat flow is the same through both**: $k_C A \frac{\Delta\theta_C}{x_C} = k_D A \frac{\Delta\theta_D}{x_D}$.
- With the same A: $k_C(d\theta/dx)_C = k_D(d\theta/dx)_D$, so the material with the **smaller k has the larger temperature gradient** (the steeper line).

Simulation — Conduction through a composite bar

Choose the two materials and their lengths, and set the end temperatures. The graph shows the steady-state temperature along the bar, the interface temperature and the rate of heat flow. Switch off the lagging to see the curve of an unlagged bar.



interface temperature	gradient in C	gradient in D	rate of heat flow (A = 1 cm²)
90.2 °C	**−74.7 °C/m**	**−140.5 °C/m**	**1.53 W**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 12 to use the simulation.

Example 12.1

A metal cube of side 8 cm has thermal conductivity 250 W m^{-1} K^{-1}. Two opposite faces are at 90 °C and 10 °C. Calculate (a) the temperature gradient in the cube, (b) the heat that flows through it in 10 minutes. (Steady flow, no heat lost to the surroundings.)

Solution

$A = (8 \times 10^{-2})^2 = 64 \times 10^{-4}$ m^2

(a) $\dfrac{d\theta}{dx} = \dfrac{10 - 90}{8 \times 10^{-2}} = \mathbf{-1.0 \times 10^3\ K\ m^{-1}}$

(b) $\dfrac{dQ}{dt} = -kA\dfrac{d\theta}{dx} = (250)(64 \times 10^{-4})(1.0 \times 10^3) = 1600$ W; $Q = 1600 \times 600 = \mathbf{9.6 \times 10^5\ J}$

Example 12.2

A 5 mm copper plate is sealed to a 10 mm aluminium plate; both have area 1 m^2. The outer face of the copper is at 100 °C and that of the aluminium at 80 °C. (a) Find the temperature of the copper–aluminium interface. (b) Calculate the rate of heat flow. (k_{Cu} = 400, k_{Al} = 200 W m^{-1} $°C^{-1}$)

Solution

(a) Same rate through both: $\frac{400(1)(100-\theta)}{5\times10^{-3}} = \frac{200(1)(\theta-80)}{10\times10^{-3}} \Rightarrow 4(100-\theta) = \theta - 80 \Rightarrow$ **θ = 96 °C**

(b) $\frac{dQ}{dt} = \frac{400(1)(100-96)}{5\times10^{-3}} =$ **3.2×10^5 W**

Exercises: conduction

1. A metal plate 5.0 cm thick has area 300 cm^2. One face is kept at 100 °C (steam) and the other at 30 °C (flowing water). The rate of heat flow is 9 kW. Find the thermal conductivity of the metal.

   Answer

   214 W m^{-1} K^{-1}

2. A 1.300 m rod is made of 0.800 m of aluminium joined end to end to 0.500 m of brass. The free end of the aluminium is at 150.0 °C and that of the brass at 20.0 °C; no heat is lost from the sides. Find the temperature of the junction. (k_{Al} = 205, k_{brass} = 109 W m^{-1} $°C^{-1}$)

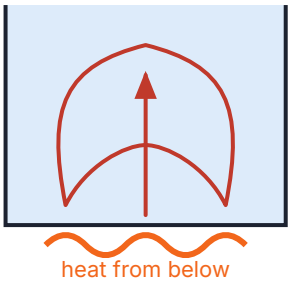
   Answer

   90.2 °C

12.2 Convection

Convection

is the process by which **heat is transferred from one part of a fluid to another by the movement of the fluid itself**. It occurs in liquids and gases.



- The fluid moves because its density changes when it is heated.
- In a pot of water heated from below, the heated water at the bottom expands, its density falls and it rises; cooler, denser water sinks to take its place. The water circulates in a **convection current**.

Natural (free) convection	Forced convection
The fluid moves because of density differences caused by heating.	The fluid is made to move by a pump or a fan.
e.g. warm and cold ocean currents, land and sea breezes, water heated in a pot	e.g. air heated and blown into a room by a fan; car radiator cooling

12.3 Radiation

Radiation

is the process by which **heat is transferred by electromagnetic waves** (infrared, visible light, ultraviolet). It can take place in a vacuum.

- All objects radiate energy continuously as electromagnetic waves produced by the thermal motion of their particles.
- The wavelengths emitted depend on the temperature: the lower the temperature, the longer the wavelength at which most energy is emitted.

Stefan–Boltzmann law

The **rate of energy radiated per unit surface area of a body is proportional to the fourth power of its absolute temperature**.

$$\frac{dQ}{dt} = P = e\sigma AT^4$$

$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} (Stefan–Boltzmann constant); A = surface area; T in kelvin; e = emissivity, $0 \le e \le 1$ (dimensionless)

- $e = 0$: ideal reflector; $e = 1$: ideal absorber (a black body). A good absorber is also a good emitter.
- An object at T_0 in surroundings at T_s also absorbs radiation from them. The **net** rate of energy radiated is $P_{net} = e\sigma A(T_0^4 - T_s^4)$.
- P_{net} positive: net heat flows out and the body cools. Negative: net heat flows in and the body warms.

Example 12.3

Tungsten has emissivity 0.35. A tungsten sphere of radius 3.00 cm hangs in a large evacuated enclosure whose walls are at 300 K. What power input keeps the sphere at 2500 K? (Ignore conduction along the supports; $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4})

Solution

$A = 4\pi r^2 = 4\pi(3.00 \times 10^{-2})^2 = 1.13 \times 10^{-2}$ m^2

$P_{net} = e\sigma A(T_0^4 - T_s^4) = (0.35)(5.67 \times 10^{-8})(1.13 \times 10^{-2})(2500^4 - 300^4) =$ **8.77 kW**

Exercise: radiation

1. (a) Find the power radiated by a tungsten sphere ($e = 0.35$) of radius 18.0 cm at 25 °C. (b) If the sphere is in a room whose walls are at −5 °C, find the net rate of energy flow out of the sphere.

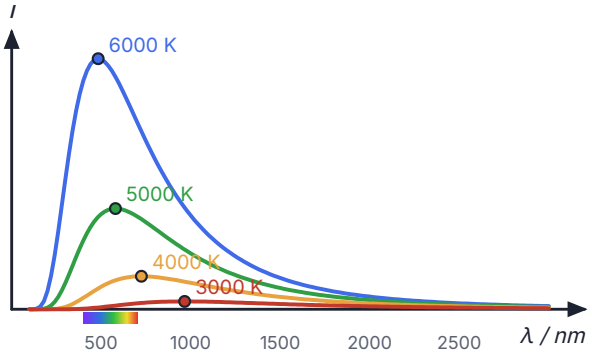
Answer

(a) 64 W (b) 22 W

Black body and its emission spectra

Black body

is an **ideal body that absorbs all the radiation incident on it** ($e = 1$); it is also the best possible emitter of thermal radiation at every wavelength.



Black-body emission spectra at four temperatures (same vertical scale).

- At a given temperature a black body emits a **continuous spectrum** over a range of wavelengths, with a peak at λ_{max}.
- As the temperature increases, the **total energy radiated** (the area under the curve) increases ($\propto T^4$).
- The **peak shifts to shorter wavelengths**, and the curve becomes taller and narrower.

Wien's displacement law

The **wavelength at which a black body emits most strongly is inversely proportional to its absolute temperature**.

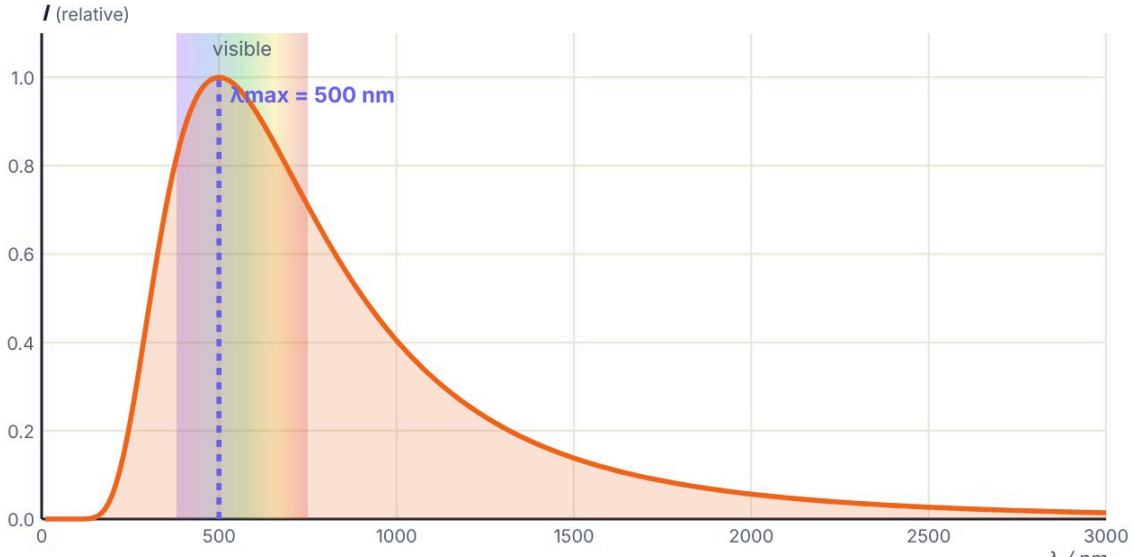
$$\lambda_{max} T = b$$

$b = 2.898 \times 10^{-3}$ m K (Wien's constant)

Simulation

Black-body spectrum and Wien's law

Change the temperature. The peak wavelength follows λmax = b/T and the total power grows as T^4. Compare a hot star, the Sun, a light bulb filament and a human body.



each curve is scaled to its own peak; the real peak height grows as T^5

temperature	λmax = b/T	power per m^2 σT^4	compared with 5800 K
5800 K	**500 nm (visible)**	**64.165 MW/m²**	**1.000 ×**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 12 to use the simulation.

Example · Wien's law

The Sun's surface is at about 5800 K. Estimate the wavelength at which it radiates most strongly.

Solution

$$\lambda_{max} = \frac{b}{T} = \frac{2.898 \times 10^{-3}}{5800} \approx \mathbf{5.0 \times 10^{-7}\ m = 500\ nm}\ \text{(visible light)}$$

12.4 Global warming

The greenhouse effect

- The Sun sends energy as light and heat. Some is reflected back into space; the rest passes through the atmosphere and warms the Earth.
- The Earth radiates heat back towards space as **infrared** radiation (longer wavelength, because the Earth is much cooler than the Sun).
- Some gases in the atmosphere (**greenhouse gases**) absorb this infrared radiation and re-emit part of it back towards the Earth, trapping heat like a blanket. This is the **natural greenhouse effect**; without it the Earth would be much colder and could not support life.

> **Global warming**
> is the increase in the average temperature of the Earth caused by the **enhanced greenhouse effect**: human activities increase the amount of greenhouse gases, so more heat is trapped and less escapes to space.

Natural greenhouse effect	Enhanced greenhouse effect
The atmosphere traps some of the heat radiated by the Earth.	More greenhouse gases trap more heat.
Some heat still escapes to space.	Less heat escapes to space.
The Earth stays warm enough for life.	The Earth heats up and the climate changes.

Greenhouse gases and human activity

- The atmosphere is 78% nitrogen and 21% oxygen, which do not trap heat. The greenhouse gases make up less than 1%: **carbon dioxide, methane, nitrous oxide, ozone, water vapour and halocarbons**.
- **Carbon dioxide**: burning fossil fuels (the largest source) and deforestation (fewer trees to absorb CO_2).
- **Methane**: livestock farming, flooded rice fields and decomposing rubbish in landfills; the amount has increased about two and a half times since the Industrial Revolution.

Ways to reduce global warming

- Switch to renewable energy: solar, wind and hydroelectric power instead of fossil fuels.
- Improve energy efficiency: energy-saving appliances, insulated buildings, less wasted electricity.
- Reduce, reuse and recycle: less waste in landfills, so less methane.
- Plant trees (reforestation and afforestation) to absorb carbon dioxide.
- Use public transport and electric vehicles to cut emissions.
- Support sustainable policies and international agreements to limit greenhouse gas emissions.